

Federal Contract Spend Analysis

A full-pipeline analysis of U.S. federal contracting data — agency spend, NAICS industry trends, and contractor tier segmentation.

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Data Source	USAspending.gov public API (DATA Act — no bulk downloads used)
Fiscal Year	FY2024 (Oct 2023 – Sep 2024)
Stack	Python · SQL (SQLite) · Jupyter Notebook · Tableau Public
GitHub	Github
Dashboard	Tableau
LinkedIn	Linkedin
Status	Complete — pipeline + dashboard published

1. Project Overview

Federal contracting data is rich, public, and underused. This project builds a full analytics pipeline — from raw API data to a business-ready dashboard — answering three core questions:

- Which federal agencies are spending the most, and how efficiently?
- Which industries (NAICS codes) are capturing the most contract value?
- What does the contractor landscape look like by size tier?

Why this matters

Federal contracting data requires the same research techniques used in government business development: company segmentation by contract behavior, NAICS-based industry profiling, and spend trend analysis. This project operationalizes those techniques on open federal data.

2. Data Source

All data is sourced from the USAspending.gov public API — no account or API key required. The data is publicly mandated under the DATA Act.

Dataset	Contents	Scope
agency_spend	Agency budgetary resources and obligations	6 agencies, FY2024

naics_awards_raw	Contract awards filtered by NAICS sector prefix	600 records across 6 sectors
award_distribution	Top agencies by total award value	Top 50 agencies

API compliance note

This project uses the USAspending.gov public API only — not bulk downloads. Use is fully compliant with the DATA Act and USAspending terms of service.

3. Pipeline Architecture

Script	Stage	Description
01_pull_data.py	Pull	Calls USAspending API, saves 3 raw CSVs to /data/. Includes retry logic and CGAC-based agency targeting.
02_clean_data.py	Clean	Validates, standardizes, engineers features: contract tiers, obligation rates, durations, state codes.
03_analysis_queries.sql	Analyze	12 standalone SQL queries covering spend, NAICS, tiers, and concentration. For reference and interview walkthroughs.
04_run_sql_analysis.py	Load+Run	Loads CSVs into SQLite, runs all queries, exports results to /output/ as CSVs and JSON.
Dashboard	Visualize	KPI cards, bar charts, treemaps, tier breakdown — Tableau or Power BI.

4. Key Findings (FY2024)

Metric	Value
Total contract spend analyzed	\$1.524 Trillion
Unique contractors	235
Average contract size	\$2.56 Billion

Tier A share of spend	100% (all awards $\geq$ \$10M)
Top agency by spend	Department of Defense — \$446B (60.2% of total)
Top agency by budget efficiency	Dept. of Health & Human Services — 87.9% obligation rate
Second largest spender	Dept. of Veterans Affairs — \$66.9B (9.0%)
DoD + VA combined share	69.2% of all federal contract spend

Key insight

The Department of Defense alone accounts for 60% of all federal contract spend — nearly 7x the next largest agency (Veterans Affairs). HHS leads on budget efficiency at 87.9%, meaning it commits nearly all allocated funds. Department of Energy has the lowest efficiency at 53.3%, signaling significant unspent obligations.

5. Key Engineering Decisions

5.1 Contractor Tier Segmentation

Awards are classified into four tiers by dollar value — mirroring industry-standard government contracting research methodology:

Tier	Award Size	Profile
A — Large	$\geq$ \$10M	Strategic partners, major prime contractors
B — Mid	\$1M–\$10M	Established contractors, subprime candidates
C — Small	\$100K–\$1M	Small business targets
D — Micro	<\$100K	Sole source / simplified acquisition

Note: All 595 awards in this pull were Tier A ($\geq$ \$10M), reflecting the large-contract focus of the targeted API endpoints.

5.2 Obligation Rate Analysis

The pipeline calculates `obligation_rate_pct` — the share of budgetary resources actually committed per agency. This surfaces agencies that under-spend their allocations, a key signal for spend efficiency analysis.

5.3 API Fixes Applied

Two endpoint corrections were required during development:

- Department of Agriculture CGAC code corrected from 021 to 012
- Award distribution endpoint updated from /spending_by_category/ to /spending_by_category/awarding_agency/

5.4 Jupyter Compatibility Fixes

Scripts were adapted for JupyterLab Desktop compatibility:

- Replaced `__file__` references with `os.getcwd()` for path resolution
- Resolved numpy/pandas import conflict in JupyterLab Desktop's bundled Python environment
- Added numpy-to-Python type serialization for JSON export compatibility

6. SQL Analysis Highlights

All 12 queries are in `scripts/03_analysis_queries.sql`. Key analytical patterns used:

Cumulative spend concentration

Window function (`SUM OVER ORDER BY`) calculates rolling market share — showing how much of total federal spend is captured by the top N agencies.

Tier A contractor identification

Filters to awards $\geq \$10\text{M}$ and ranks by recipient, surfacing major prime contractors per agency and sector.

NAICS by state cross-tab

Groups awards by `state_code` and `naics_sector` to reveal geographic concentrations of industry activity.

7. How to Run

Step 1: Install dependencies

```
pip install requests pandas numpy
```

Step 2: Pull data from USAspending API

```
python scripts/01_pull_data.py
```

Step 3: Clean and validate

```
python scripts/02_clean_data.py
```

Step 4: Run SQL analysis

```
python scripts/04_run_sql_analysis.py
```

Step 5: Open output CSVs in Tableau or Power BI to build the dashboard.

Jupyter users

Replace `os.path.dirname(__file__)` with `os.getcwd()` in all path references. Use `sys.executable -m pip install` to install packages into the correct environment.

8. Skills Demonstrated

Skill	Evidence
Data Collection	REST API integration, pagination, retry logic, CGAC-based agency targeting
Python	Pandas, NumPy, data type coercion, feature engineering, JSON export
SQL	Aggregations, window functions, cross-tabulations, KPI queries, SQLite
Data Quality	Validation framework, null handling, deduplication, type standardization
BI & Dashboarding	KPI card design, multi-view layout, filter architecture (in progress)
API Compliance	DATA Act public API usage, no bulk downloads, proper attribution
Documentation	README, pipeline architecture, business-framed findings, GitHub repo

9. Dashboard

Published live on Tableau Public. Three views:

View	Description
Agency Spend	Top 10 agencies by total contract spend (\$B), colored by market share
Obligation Efficiency	Agency budget efficiency rates with 80% threshold line, color-coded High/Moderate/Low

Tier Distribution	All 595 contracts are Tier A ($\geq$ \$10M), totaling \$1.524T in contract value
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Live dashboard linkpublic.tableau.com/views/FederalContractSpendAnalysisFY2024/Dashboard1

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